

2D SCANNING LIDAR · LR-1BS SERIES

SINGLE-OUTPUT OLEI LR-1BS2H 270° LIDAR

The same 270° scanner and the same 0.2–10 m envelope as the LR-1BS2, wired for one job: one switching output, separate M8 power and LAN leads, and a point cloud on Ethernet for everything else.



■ WHY THIS SENSOR

- **270° field of view from one corner mount** — front and both flanks of a platform covered by a single sensor
- **One switching output** — a clean stop or warn contact where the platform only needs one decision
- **Separate M8 power and LAN connectors** — 5-pin M8 B-code for power, 4-pin M8 A-code for Ethernet — no breakout harness
- **Point cloud over 100 Mbps Ethernet** — azimuth, distance and signal strength alongside the switching output
- **0.225° angle resolution at 10–25 Hz** — 16K–40K points per second, 130 ms response at 10 Hz
- **≤ ±1° horizontal plane error** — the scan plane stays where the drawing says it does
- **5 W at 12–24 V DC, IP65, 450 g** — platform power, wash-down tolerant, no fan
- **OLAMViewer and a ROS driver** — stream into an existing perception stack on day one

0.2–10 m

WORKING RANGE

270°

FIELD OF VIEW

130 ms

RESPONSE @ 10 HZ

450 g

WEIGHT

■ APPLICATION SCENARIOS



AMR & AGV

270° obstacle avoidance from a single chassis-mounted scanner



INTRALOGISTICS

Forklifts, tuggers and conveyor zones in active aisles



INDUSTRIAL CELLS

Machine-envelope monitoring and presence sensing



AREA MONITORING

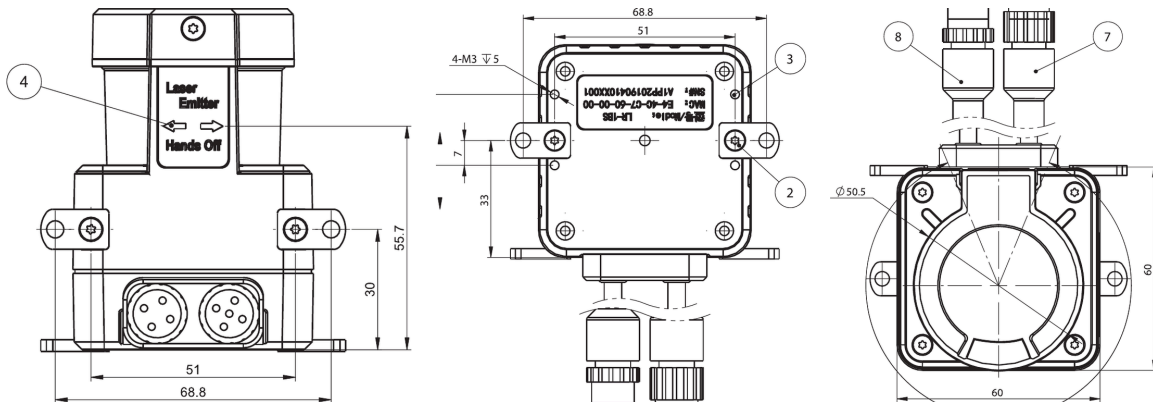
Perimeter and equipment-envelope intrusion detection

LR-1BS2H — SPECIFICATIONS

PARAMETER	TECHNICAL DATA
Sensing technology	2D scanning ToF LiDAR
Laser wavelength	905 ± 20 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Laser channels	1
Field of view (horizontal)	270°
Beam divergence	5.8 mrad longitudinal 0.58 mrad transversal
Angle resolution (horizontal)	0.225°
Scanning frequency	10–25 Hz
Working distance (based on reflectivity)	0.2–10 m @ 80% reflectivity 0.2–8 m @ 10% · 0.2–2 m @ 4% · 0.2–10 m @ reflector
Accuracy ①	Absolute < ±50 mm Repeat < 20 mm
Distance resolution	1 mm
Signal strength	0–60000
Scanning rate	16K–40K pts/s
Horizontal error (horizontal angle)	≤ ±1°
Response time	130 ms @ 10 Hz

PARAMETER	TECHNICAL DATA
Switching outputs	1
Ethernet connection	100 Mbps Ethernet
Connection	5-pin M8×1 B-code female, 2 m power 4-pin M8×1 A-code female, 2 m LAN
Operating voltage	12–24 V DC
Power consumption @ 25 °C	5 W
Mechanical dimensions	≤ 60 × 60 × 85 mm
Weight	450 g
Degree of protection	IP65 or optional
Operating temperature	–10 to +50 °C
Storage temperature	–30 to +70 °C
Ambient light limit	80 000 lux
Indicator status	RGB × 4
Mounting	Bottom / back
Function software	Oforge
Basic software	OLAMViewer · ROS driver

OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

Certification. FCC, EMC, CE, RoHS, REACH, FDA. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. ① Standard value; actual value depends on environmental conditions. Accuracy and working distance both degrade with lower target reflectivity — size the zones from the 4% and 10% figures, not the 80% figure. Where a platform needs multiple selectable banks and four discrete outputs, specify the LR-1BS2 instead.

Installation. Mount the scanner so the 270° plane is unobstructed through its full sweep and the blind 90° faces structure, not traffic. Keep the optical window clear of spray and dust, clean it on a schedule in dirty environments, and verify the output behaviour on the platform after any change to mounting height or angle.

OLEI and the OLEI logo are trademarks of Hangzhou OLE-Systems Co., Ltd. MorpheusTEK is the exclusive North American distributor for OLEI LiDAR. Specifications are subject to change without notice; confirm current values with MorpheusTEK before design-in. Doc. MTK-OLEI-LR1BS2H-2026.09 · Source: OLEI LR-1BS2H Mini LiDAR Sensor datasheet, DOC.# DSEN-1BS2H-202507.